

Q1 - 24 June - Shift 1

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In an uniform magnetic field, speed and energy remains the same for a moving charged particle.

Reason (R) : Moving charged particle experiences magnetic force perpendicular to its direction of motion.

(A) Both (A) and (R) are true and (R) is the correct

explanation of (A)

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true.

Space for your notes:

Q2 - 24 June - Shift 1

The magnetic field at the centre of a circular coil of radius r , due to current I flowing through it, is B .

The magnetic field at a point along the axis at a distance $\frac{r}{2}$ from the centre is :

(A) $B/2$

(B) $2B$

(C) $\left(\frac{2}{\sqrt{5}}\right)^3 B$

(D) $\left(\frac{2}{\sqrt{3}}\right)^3 B$

Space for your notes:

Q3 - 24 June - Shift 2

#MathBoleTohMathonGo

Questions

MathonGo

A proton, a deuteron and an α -particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is :

(A) $1:\sqrt{2}:\sqrt{2}$ (B) $1:1:\sqrt{2}$

(C) $\sqrt{2}:1:1$ (D) $1:\sqrt{2}:1$

*Space for your notes:***Q4 - 25 June - Shift 1**

A long straight wire with a circular cross-section having radius R , is carrying a steady current I . The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance r ($r < R$) from its centre will be :-

(A) $B \propto r^2$ (B) $B \propto r$

(C) $B \propto \frac{1}{r^2}$ (D) $B \propto \frac{1}{r}$

*Space for your notes:***Q5 - 25 June - Shift 2**

A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to

(A) B (B) $2B$

(C) $4B$ (D) $\frac{B}{2}$

*Space for your notes:***Q6 - 26 June - Shift 1**

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Questions

MathonGo

A proton and an alpha particle of the same enter in a uniform magnetic field which is acting perpendicular to their direction of motion. The ratio of the circular paths described by the alpha particle and proton is:

- (A) 1 : 4 (B) 4 : 1
(C) 2 : 1 (D) 1 : 2

Q7 - 26 June - Shift 2

A bar magnet having a magnetic moment of $2.0 \times 10^5 \text{ JT}^{-1}$, is placed along the direction of uniform magnetic field of magnitude $B = 14 \times 10^{-5} \text{ T}$. The work done in rotating the magnet slowly through 60° from the direction of field is :

- (A) 14 J (B) 8.4 J (C) 4 J (D) 1.4 J

Q8 - 26 June - Shift 2

A metal surface is illuminated by a radiation of wavelength $4500 \text{ \AA}$. The ejected photo-electron enters a constant magnetic field of 2 mT making an angle of 90° with the magnetic field. If it starts revolving in a circular path of radius 2 mm , the work function of the metal is approximately :

- (A) 1.36 eV (B) 1.69 eV (C) 2.78 eV (D) 2.23 eV

Q9 - 26 June - Shift 2

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Questions

MathonGo

Two 10 cm long, straight wires, each carrying a current of 5A are kept parallel to each other. If each wire experienced a force of 10^{-5} N, then separation between the wires is _____ cm.

*Space for your notes:***Q10 - 27 June - Shift 2**

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Questions

MathonGo

Two long parallel conductors S_1 and S_2 are separated by a distance 10 cm and carrying currents of 4A and 2A respectively. The conductors are placed along x-axis in X-Y plane.

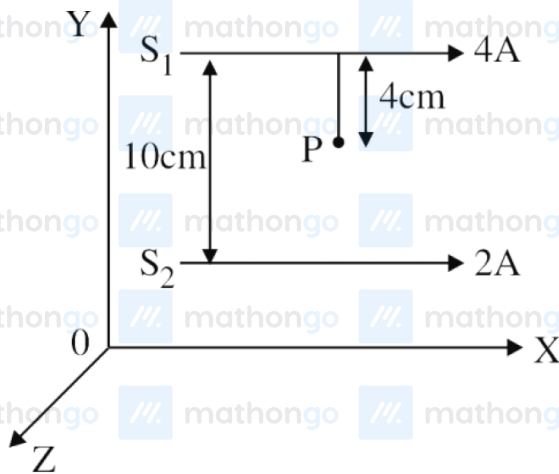
There is a point P located between the conductors (as shown in figure).

A charge particle of 3π coulomb is passing through the point P with velocity

$\vec{v} = (2\hat{i} + 3\hat{j})\text{ m/s}$; where $\hat{i}$ & $\hat{j}$ represents unit

vector along x & y axis respectively.

The force acting on the charge particle is $4\pi \times 10^{-5}(-x\hat{i} + 2\hat{j})\text{ N}$. The value of x is :



(A) 2

(B) 1

(C) 3

(D) -3

Space for your notes:

Q11 - 27 June - Shift 2

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Questions

MathonGo

A deuteron and a proton moving with equal kinetic energy enter into to a uniform magnetic field at right angle to the field. If r_d and r_p are the radii of their circular paths respectively, then the ratio $\frac{r_d}{r_p}$ will be $\sqrt{x} : 1$ where x is _____.

*Space for your notes:***Q12 - 28 June - Shift 1**

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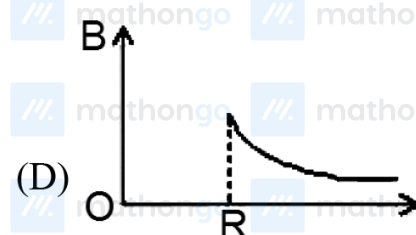
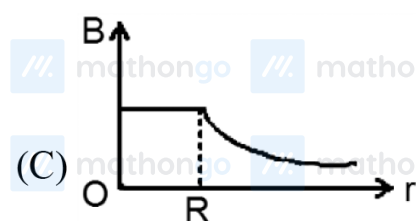
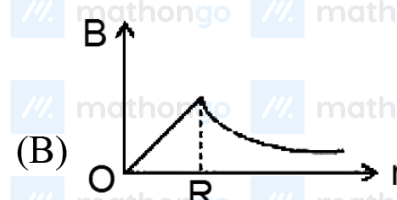
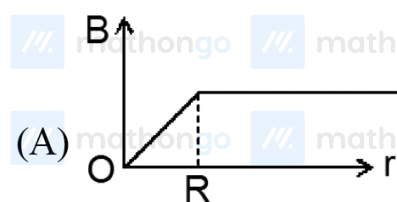
Questions

MathonGo

An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface.

Space for your notes:

Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.



Q13 - 28 June - Shift 1

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A singly ionized magnesium atom (A_{24}) ion is accelerated to kinetic energy 5 keV and is projected perpendicularly into a magnetic field B of the magnitude 0.5 T. The radius of path formed will be _____ cm.

Space for your notes:

Q14 - 28 June - Shift 2

Two parallel, long wires are kept 0.20 m apart in vacuum, each carrying current of x A in the same direction. If the force of attraction per meter of each wire is 2×10^{-6} N, then the value of x is approximately:

Space for your notes:

(A) 1 (B) 2.4

(C) 1.4 (D) 2

Q15 - 28 June - Shift 2

Questions

MathonGo

A coil is placed in a time varying magnetic field. If the number of turns in the coil were to be halved and the radius of wire doubled, the electrical power dissipated due to the current induced in the coil would be:

(Assume the coil to be short circuited.)

(A) Halved

(B) Quadrupled

(C) The same

(D) Doubled

Space for your notes:

Q16 - 29 June - Shift 1

A charged particle moves along circular path in a uniform magnetic field in a cyclotron. The kinetic energy of the charged particle increases to 4 times its initial value. What will be the ratio of new radius to the original radius of circular path of the charged particle :

(A) 1 : 1

(B) 1 : 2

(C) 2 : 1

(D) 1 : 4

Space for your notes:

Q17 - 29 June - Shift 2

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Questions

MathonGo

Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is $300 \mu\text{T}$. The equal current flowing in the two conductors is :

- (A) 30A in the same direction.
- (B) 30A in the opposite direction.
- (C) 60A in the opposite direction.
- (D) 300A in the opposite direction.

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Answer Key

Q1 (A)

Q2 (C)

Q3 (D)

Q4 (B)

Q5 (A)

Q6 (C)

Q7 (A)

Q8 (A)

Q9 (5)

Q10 (C)

Q11 (2)

Q12 (D)

Q13 (10)

Q14 (C)

Q15 (B)

Q16 (C)

Q17 (B)

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Q1 (A)

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\vec{F} \perp \vec{v}$$

$$\text{Work done} = \vec{F} \cdot \vec{S}$$

$$\text{Work done} = 0$$

Q2 (C)

$$B_c = \frac{\mu_0 I}{2r}, B_a = \frac{\mu_0 I r^2}{2(x^2 + r^2)^{3/2}}$$

$$\text{At } x = \frac{r}{2}$$

$$B_a = \frac{\mu_0 I r^2}{2 \left(\frac{r^2}{4} + r^2 \right)^{3/2}}$$

$$= \frac{\mu_0 I r^2}{2 \left(\frac{5}{4} r^2 \right)^{3/2}} = \frac{\mu_0 I}{2r} \left(\frac{4}{5} \right)^{3/2}$$

$$= \frac{\mu_0 I}{2r} \left(\frac{2}{\sqrt{5}} \right)^3$$

Q3 (D)

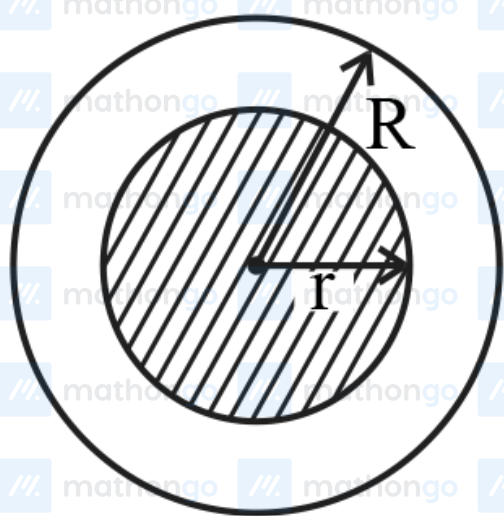
$$R = \frac{\sqrt{2km}}{qB} \propto \frac{\sqrt{m}}{q}$$

$$\frac{\sqrt{m}}{e} : \frac{\sqrt{2m}}{e} : \frac{\sqrt{4m}}{2e}$$

$$1 : \sqrt{2} : 1$$

Q4 (B)

Use Ampere's law



$$B \cdot 2\pi r = \mu_0 \cdot \frac{I}{\pi R^2} \cdot \pi r^2$$

$$\text{Thus } B \propto r$$

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Q5 (A)

$$B_1 = \mu_0 n I$$

$$B_2 = \mu_0 \left(\frac{n}{2} \right) (2I)$$

$$\Rightarrow B_1 = B_2$$

Q6 (C)

$$\frac{R_\alpha}{R_P} = \frac{M_\alpha}{M_P} \times \frac{q_P}{q_\alpha}$$

$$\frac{R_\alpha}{R_P} = \frac{4}{1} \times \frac{1}{2} = 2$$

Q7 (A)

$$\text{Work done} = MB (\cos \theta_1 - \cos \theta_2)$$

$$\theta_1 = 0^\circ, \theta_2 = 60^\circ$$

$$= 2 \times 10^5 \times 14 \times 10^{-5} (1 - 1/2)$$

$$= 14 \text{ J}$$

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Q8 (A)

$$\lambda = 4500 \text{ \AA}$$

$$B = 2 \text{ mT}, R = 2 \text{ mm}$$

$$R = \frac{\sqrt{2Km}}{qB}$$

$$\frac{(qBR)^2}{2m} = K$$

$$\frac{(1.6 \times 10^{-19} \times 2 \times 10^{-3} \times 2 \times 10^{-3})^2}{2 \times 9.1 \times 10^{-31}} = K$$

$$\frac{(6.4)^2}{2 \times 9.1} \times \frac{10^{-50}}{10^{-31}} = K$$

$$K = 2.25 \times 10^{-19} \text{ J}$$

$$= \frac{2.25 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} = 1.40 \text{ eV}$$

$$E = \frac{12400}{4500} = 2.76 \text{ eV}$$

$$\phi = E - K = (2.76 - 1.40) \text{ eV} = 1.36 \text{ eV}$$

Q9 (5)

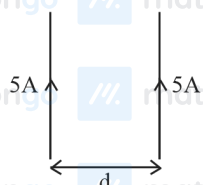
Hints and Solutions

MathonGo

It should be mentioned, 10 cm wire is part of long wire.

Force experienced by unit length of wire

$$= \frac{\mu_0 I_1 I_2}{2\pi d}, I_1 = I_2 = 5A$$



Force experienced by wires of length 10 cm

$$= \frac{\mu_0 I_1 I_2}{2\pi d} \times 10 \times 10^{-2}$$

$$10^{-5} = \frac{2 \times 10^{-7} \times 5 \times 5}{d} \times 10 \times 10^{-2}$$

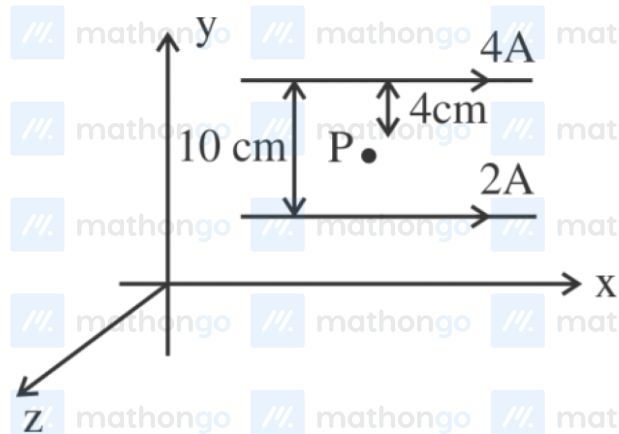
$$d = 50 \times 10^{-3} \text{ m}$$

$$d = 50 \times 10^{-1} \text{ cm} = 5 \text{ cm.}$$

Q10 (C)

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$$B_{\text{net}} = B_1 - B_2 = \frac{\mu_0 \times 4}{2\pi[.04]} - \frac{\mu_0 \times 2}{2\pi[.06]}$$

$$\vec{B}_{\text{net}} = \frac{\mu_0}{2\pi} \left[\frac{200}{3} \right] (-\hat{k})$$

$$\vec{F} = q[\vec{v} \times \vec{B}]$$

$$= [3\pi] \left[(2\hat{i} + 3\hat{j}) \times \left(\frac{\mu_0}{2\pi} \right) \left(\frac{200}{3} \right) - \hat{k} \right]$$

$$= 3\pi \times \frac{\mu_0}{2\pi} \left(\frac{200}{3} \right) [2 \times \hat{j} - 3(\hat{i})]$$

$$= (4\pi \times 10^{-7})(100)(-3\hat{i} + 2\hat{j})$$

$$= 4\pi \times 10^{-5} \times [-3\hat{i} + 2\hat{j}]$$

Q11 (2)



$$R = \frac{mv}{qB}$$

$$R_D = \frac{(2m_p) v_D}{eB}$$

$$R_P = \frac{(m_p) v_P}{eB}$$

$$\frac{R_D}{R_P} = \frac{2v_D}{v_P} = \frac{2v_D}{\sqrt{2}v_D} = \frac{\sqrt{2}}{1}$$

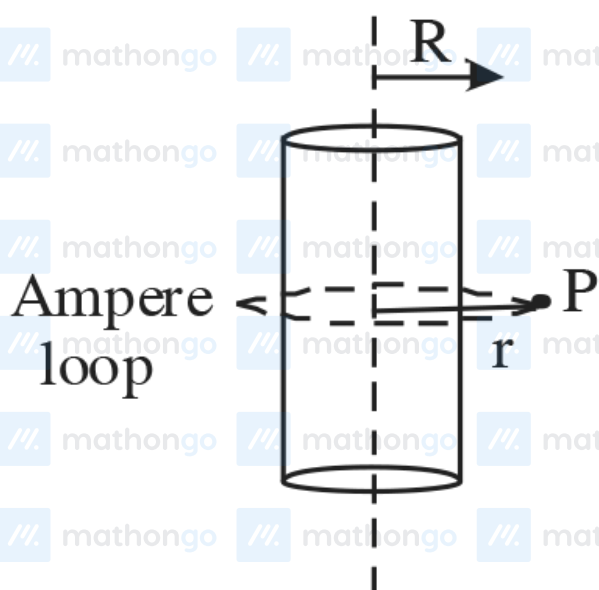
$$\frac{1}{2}(2m_p)v_D^2 = \frac{1}{2}m_p \cdot v_P^2$$

$$\sqrt{2}v_D = v_P$$

$$x = 2$$

#MathBoleTohMathonGo

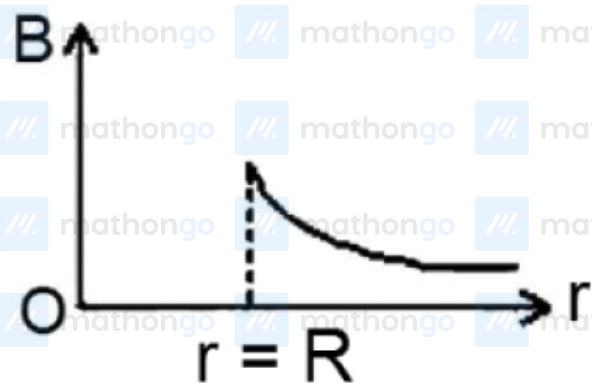
Q12 (D)



$$1) r < R, B_p = 0$$

$$2) r \geq R, B_p = \frac{\mu_0 I}{2\pi r}$$

$$B_p \propto \frac{1}{r}$$



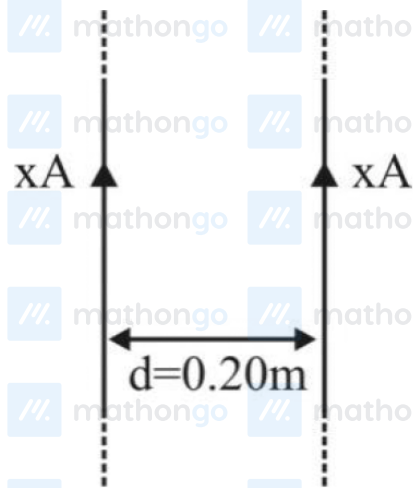
#MathBoleTohMathonGo

Q13 (10)

$$R = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

Q14 (C)

#MathBoleTohMathonGo



$$\text{Force per unit length} = \frac{\mu_0 i_1 i_2}{2\pi d}$$

$$= \frac{\mu_0 \cdot x^2}{2\pi \times 0.2}$$

$$F = 2 \times 10^{-6} = \frac{4\pi \times 10^{-7} \times x^2}{2\pi \times 0.2}$$

$$\Rightarrow 10^{-6} = 10^{-7} \frac{x^2}{0.2}$$

$$\Rightarrow x^2 = 10 \times 0.2$$

$$= 2$$

$$\Rightarrow x = \sqrt{2} \approx 1.4 \text{ Amp.}$$

Q15 (B)

$$P = \frac{\varepsilon^2}{R} = \frac{\left(NA \frac{dB}{dt}\right)^2 \times A_c}{\rho \ell}$$

$$P' = \frac{\left(\frac{NA}{2} \frac{dB}{dt}\right)^2 \times 4A_c}{\rho \ell / 2}$$

$$\Rightarrow P' = 2P$$

Q16 (C)

radius of particle in cyclotron

$$r = \frac{\sqrt{2mK.E.}}{qB}$$

So ratio of new radius to original

$$\frac{r_n}{r_0} = \sqrt{\frac{(K.E.)_n}{(K.E.)_0}} = \sqrt{4} \Rightarrow 2:1 \text{ (C)}$$

Q17 (B)



$$B \text{ at } O = 2 \frac{\mu_0 I}{2\pi r}$$

$$\frac{2 \times 4\pi \times 10^{-7} I}{2\pi \times 4 \times 10^{-2}} = 3 \times 10^{-4} \text{ T}$$

$I = 30 \text{ A}$ in opp. direction